

Time Series Models: From Statistics to AI

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Exercise Sheet 5

Regulations

Please submit your solutions via Moodle in teams of **2 students**, before the exercise group on **Wednesday, May 28th, 2025**. Each submission must include **exactly** two files:

- A .pdf file containing both your Jupyter notebook and solutions to analytical exercises. The Jupyter notebook can be exported to pdf by selecting **File** → **Download as** → **pdf** in JupyterLab. If this method does not work, you may print the notebook as a pdf instead. Your analytical solutions can be either scanned handwritten solutions or created using $\text{\LaTeX}$.
- A .ipynb file containing your code as Jupyter notebook.

Both files must follow the naming convention:

Lastname1-Lastname2-sheet05.pdf

Lastname1-Lastname2-sheet05.ipynb

1 Fixed Points, Stability, and Bifurcations

Consider the univariate nonlinear map:

$$x_{t+1} = f(x_t, w, \theta) = w \cdot \sigma(x_t) + \theta, \quad \text{with} \quad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

1. For $w = 7$ and $\theta = -2.5$, find the fixed points of the system. Visualize these in a graph. Are they stable?
2. For $w = 7$, plot the bifurcation graph as a function of $\theta \in [-10, 0]$. Include both stable and unstable objects. How does the system change its dynamical properties as θ is varied within this range?

2 The Logistic Map

Consider the map:

$$x_{n+1} = f(x_n) = rx_n(1 - x_n)$$

1. Prove that, for $0 \leq x_n \leq 1$ and $0 \leq r \leq 4$, we have $0 \leq f(x_t) \leq 1$.

2. Given a trajectory $x_1, x_2, \dots, x_T$, here's a recipe to make a cobweb plot of a map: 1. Plot (x_1, x_2) . 2. Connect that point to (x_2, x_2) . 3. Connect this to (x_2, x_3) , and so forth. Starting from some $x_1 \in (0, 1)$, plot a cobweb plot for $r = 0.5, 1.5, 2.5, 3.5$ and 3.9 with 30 steps.
3. Show that the logistic map has a cycle of order 2 for $r > 3$. Use that a 2-cycle requires $f(q) = p$ and $f(p) = q$. What is stability of the 2-cycle? Does the stability change for some $r > 3$?
4. For each $r \in \{0.001, 0.002, 0.003 \dots 3.998, 3.999\}$, produce 1000 trajectories starting at random initial conditions with 100 steps. Plot only the endpoints of the trajectories according to their respective r in a 2D-scatter plot. (Hint: Since you only need to plot the last points of the trajectories, don't store the trajectories in order to save memory. Hint 2: Make your code flexible so you can use it with other maps as well).
5. Redo the plot for $r \in \{3.44500, 3.44501, 3.44502, \dots, 3.56999, 3.57000\}$ and use it to find another r such that the logistic map has a cycle of order $p \geq 3$. Then show the existence of the cycle mathematically.
6. The qualitative changes in the dynamics that you can see in the diagram for certain values of r are called bifurcations. This is where the topology of the dynamics change. Develop a procedure to numerically compute the r values for the following m bifurcations, starting from $r = 3.44$. Describe the algorithm. (Hint 1: you might have to increase floating-point precision. Hint 2: Make your algorithm independent of the map, then you can reuse it. Hint 3: This algorithm may take a long time to run, depending on your hardware. Let it run for a time that's feasible for you and see how many bifurcations you can find. That implies your code has to save a bifurcation point as soon as it finds it, because you will interrupt the code in the end.)
7. Let $2 \leq r_n \leq m - 1$ be a bifurcation point. Calculate the relative distance between successive bifurcations points

$$\delta = \frac{r_n - r_{n-1}}{r_{n+1} - r_n}$$

for all $n \in \{1, \dots, 100\}$. What do you find?

8. Consider the map:

$$x_{n+1} = r \sin(\pi x_n)$$

Compute the first bifurcations again for $r > 0.82$ and calculate δ from the exercise above. Make a bifurcation diagram (as with the logistic map) and see what it looks like.